HIGHER DIRECT IMAGES OF LQC FBC MODULES HAVE THE FLAT BASE CHANGE PROPERTY

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file stacks-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted span cites the lemmas recorded in assembly.external_references with canonical tags as navigation pointers.

Allowed standard prerequisites (editorial).

- General pushforward criterion. Cohomology of Algebraic Stacks, Lemma stacks-cohomology-lemma-general-pushforward. Supplies the four conditions whose verification yields the theorem.
- Comparison of cohomology for representable morphisms. Sheaves on Stacks, Lemma stacks-sheaves-lemma-compare-representable-morphism-cohomology. Identifies R^i f_* F with the pushforward of the restriction to the scheme cover.
- Flat base change for cohomology of algebraic spaces. Cohomology of Spaces, Lemma spaces-cohomology-lemma-flat-base-change-cohomology. Gives the isomorphism of the comparison map in the scheme case.

Source reference. Original lemma 076Z: sources/stacks-chow/tags.txt, source lines 9184–9184. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 06WL: sources/stacks-chow/tags.txt, source lines 8820–8820. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 06WM: sources/stacks-chow/tags.txt, source lines 8821–8821. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 075Y: sources/stacks-chow/tags.txt, source lines 9147–9147. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0764: sources/stacks-chow/tags.txt, source lines 9154–9154. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 075P: sources/stacks-chow/tags.txt, source lines 9138–9138. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 073K: sources/stacks-chow/tags.txt, source lines 9064–9064. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Date: stacks-project:stacks-cohomology.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 075Z: sources/stacks-chow/tags.txt, source lines 9148–9148. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let $f : \mathcal{X} \to \mathcal{Y}$ be a quasi-compact and quasi-separated morphism of algebraic stacks. Let $\mathcal{F}$ be an object of $Mod(\mathcal{X}_{étale}, \mathcal{O}_{\mathcal{X}})$ which is locally quasi-coherent and has the flat base change property. Then each $R^i f_* \mathcal{F}$ (computed in the étale topology) has the flat base change property.*

Proof. We will use Lemma 076Z to prove this. For every algebraic stack $\mathcal{X}$ let $LQCoh^{fbc}(\mathcal{O}_{\mathcal{X}})$ denote the full subcategory of $Mod(\mathcal{X}_{étale}, \mathcal{O}_{\mathcal{X}})$ consisting of locally quasi-coherent sheaves with the flat base change property. Once we verify conditions (1) – (4) of Lemma 076Z the lemma will follow. Properties (1), (2), and (3) follow from Sheaves on Stacks, Lemmas 06WL and 06WM and Lemmas 075Y and 0764. Thus it suffices to show part (4).

Suppose $f : \mathcal{X} \to \mathcal{Y}$ is a morphism of algebraic stacks such that $\mathcal{X}$ and $\mathcal{Y}$ are representable by affine schemes X and Y. In this case, suppose that $\psi : y \to y'$ is a morphism of $\mathcal{Y}$ lying over a flat morphism $b : V \to V'$ of schemes. For clarity denote $\mathcal{V} = (Sch/V)_{fppf}$ and $\mathcal{V}' = (Sch/V')_{fppf}$ the corresponding algebraic stacks. Consider the diagram of algebraic stacks

$$\begin{array}{ccccc} \mathcal{Z} & \xrightarrow{a} & \mathcal{Z}' & \xrightarrow{x'} & \mathcal{X} \\ {\scriptstyle f''}\downarrow & & {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ \mathcal{V} & \xrightarrow{b} & \mathcal{V}' & \xrightarrow{y'} & \mathcal{Y} \end{array}$$

with both squares cartesian. As f is representable by schemes (and quasi-compact and separated – even affine) we see that $\mathcal{Z}$ and $\mathcal{Z}'$ are representable by schemes Z and Z' and in fact $Z = V \times_{V'} Z'$. Since $\mathcal{F}$ has the flat base change property we see that

$$a^*_{small}\big(\mathcal{F}|_{Z'_{étale}}\big) \longrightarrow \mathcal{F}|_{Z_{étale}}$$

is an isomorphism. Moreover,

$$R^i f_* \mathcal{F}|_{V'_{étale}} = R^i(f')_{small,*}\big(\mathcal{F}|_{Z'_{étale}}\big)$$

and

$$R^i f_* \mathcal{F}|_{V_{étale}} = R^i(f'')_{small,*}\big(\mathcal{F}|_{Z_{étale}}\big)$$

by Sheaves on Stacks, Lemma 075P. Hence we see that the comparison map

$$c_\psi : b^*_{small}(R^i f_* \mathcal{F}|_{V'_{étale}}) \longrightarrow R^i f_* \mathcal{F}|_{V_{étale}}$$

is an isomorphism by Cohomology of Spaces, Lemma 073K. Thus $R^i f_* \mathcal{F}$ has the flat base change property. Since $R^i f_* \mathcal{F}$ is locally quasi-coherent by Lemma 075Z we win. □